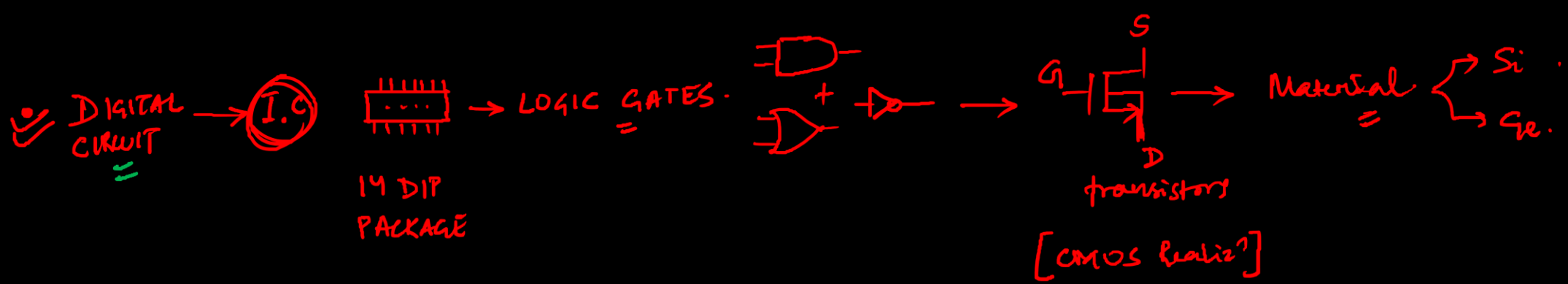


SEMICONDUCTOR PHYSICS

[RECAP]

[04-02-2025]

Sailesh Mishra



SEMICONDUCTOR PHYSICS

[02-02-2025]

Sailesh Mishra

SEMI-CONDUCTOR

INSULATORS < Conductivity < CONDUCTORS

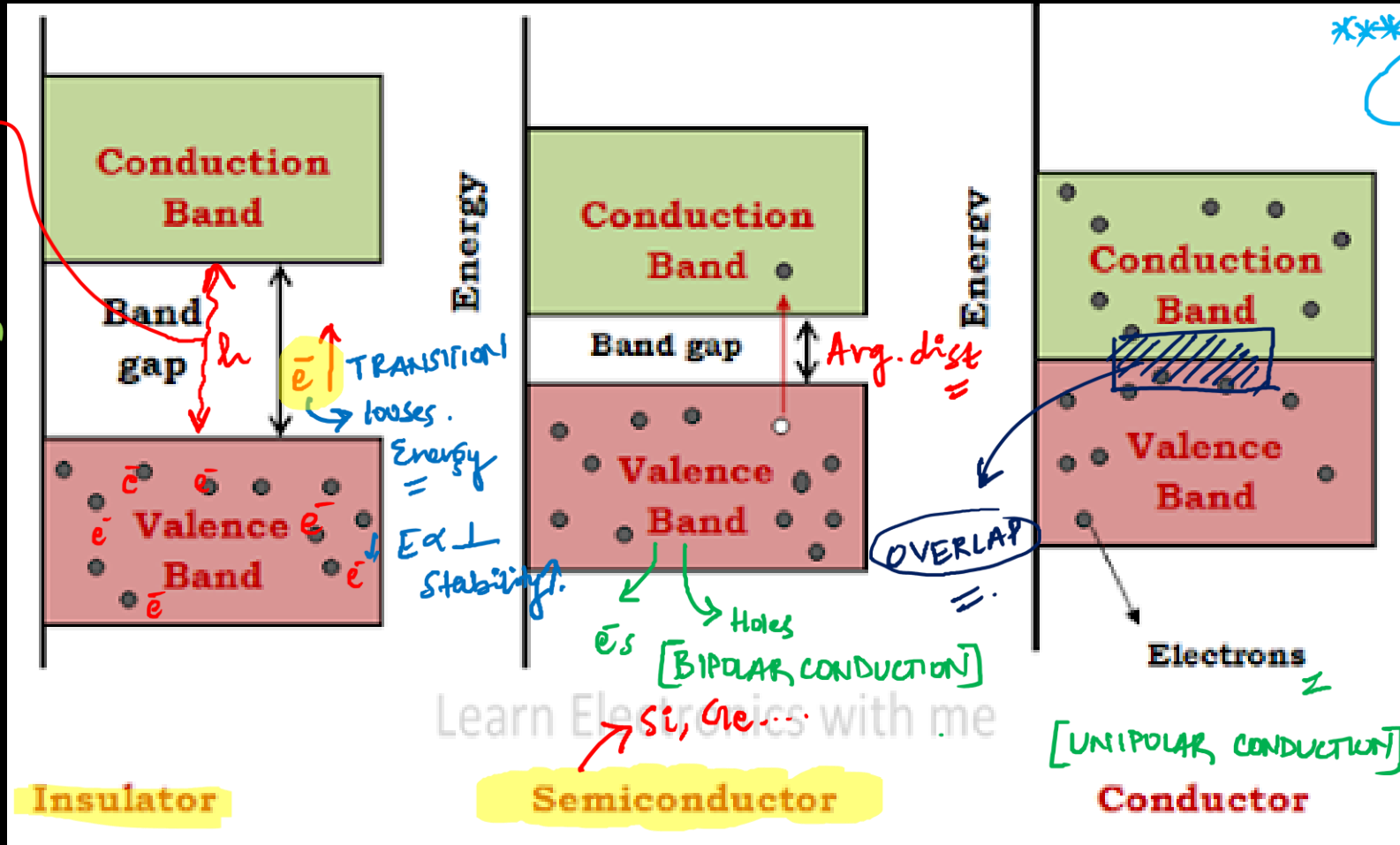
∴ This h is a distance; is a Material dependent property !!

of $h \uparrow$ PE $\uparrow$ TE $\uparrow$ (E $\uparrow$, stability $\downarrow$)

*** dist $\uparrow$ E $\uparrow$ Cond. $\downarrow$

$$E_g = \frac{1.38 \times 10^{-24} \text{ J/K} \times 300 \text{ K}}{1.6 \times 10^{-19}}$$

$$E_g = 2.5875 \times 10^{-3} \approx 25.87 \text{ mV} \approx \boxed{26 \text{ mV}}$$

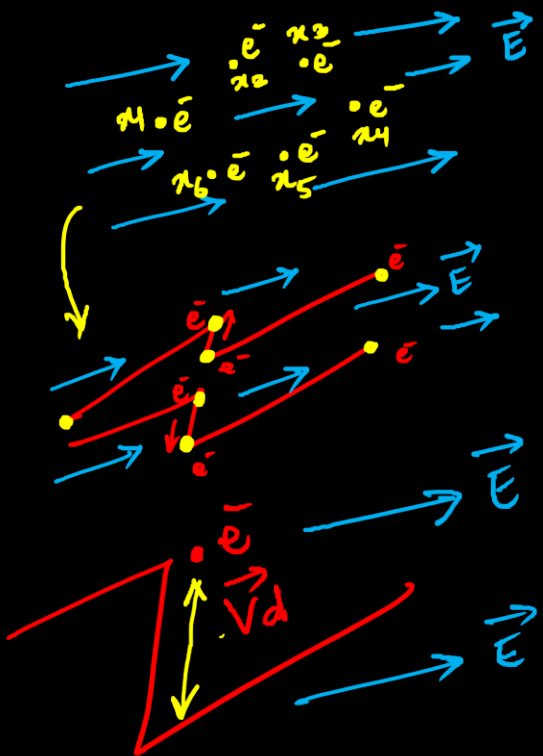


DRIFT VELOCITY = $\frac{1}{2} m v_d^2$

distance $\uparrow$ TE = PE + KE $\uparrow$ mgh
The Energy required to Make a transition from VB $\rightarrow$ CB is w/as BGEL.
[Band Gap Energy level]

• It is directly proportional to $\frac{k_B T}{2}$

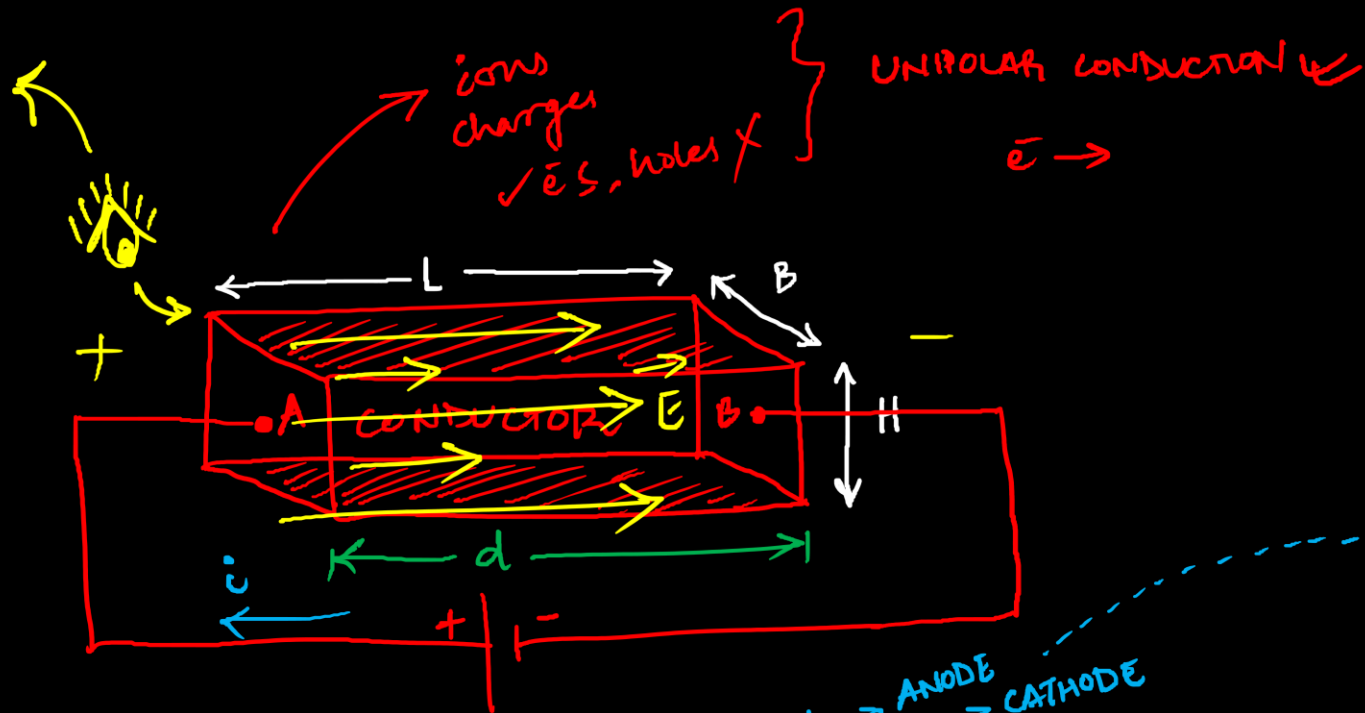
where $k_B \rightarrow$ Boltzman Constant, $T =$ Absolute Temp. = $27^\circ\text{C} = 273 + 27^\circ\text{C} = 300\text{K}$.
 $= 1.38 \times 10^{-24} \text{ J/K}$ $2 =$ charge of $e^- = 1.6 \times 10^{-19} \text{ C}$



$\vec{F} = q\vec{E}$
 $= 1.6 \times 10^{-19} \text{ C} \times 31.4 \text{ N/Cm}^2$

BATTERY
 emf (potential): ENERGY $\uparrow$
 $TE = PE + KE$

$\rightarrow$ V-I Characteristics.
 $\rightarrow$ due to emf; We're getting a $\neq \vec{E}$.



$V_{DC} = V_{AB} = V_A - V_B$: Relative Potential difference
 (9V)
 $= 9 - 0 \text{ V}$
 $= 9 \text{ V}$

$18 - 9 = 9 \text{ V} !!$
 ANODE $\rightarrow$ CATHODE

"9V Battery"



Electrolytes
 [ions, charges]
 $=$
 emf (V) $\rightarrow \vec{E}$

$\odot \vec{E} = \text{ENERGY} = \frac{1}{2} \epsilon_0 \cdot E^2 = \text{Sum of } \underline{PE} + \underline{KE} = \underline{Mgh} + \frac{1}{2} m v_d^2 = m_e [gh + \frac{1}{2} \frac{V_d^2}{f}]$

$(9.1 \times 10^{-31} \text{ Kg})$

Environment dependent

permittivity of FREE SPACE

ϵ_0 : free space or Vacuum

$= 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$

$\vec{E} = \frac{V_{dc}}{d} \propto \frac{1}{2} \epsilon_0 \cdot E^2$

$\rightarrow [V_{D.C} \propto \text{Energy of } \vec{E}]$

$\vec{E}_{coulomb} = \frac{K \cdot Q_1 Q_2}{r^2} = \frac{1}{4\pi \epsilon_0} \cdot \frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi \epsilon_0 \cdot \epsilon_r} \cdot \frac{Q_1 Q_2}{r^2}$

Relative permittivity =

Air = 1 ✓

Minerals = 3 ✓

G.C = 3-5 ✓

H₂O = 79.6 $\approx$ 80 ✓

$\rightarrow \tau$ (Relaxation Time) $\frac{0}{0}$ It's the time B/W any 2 successive collisions of e^- s inside the conductor...

$\vec{E} \uparrow \quad V_d \uparrow$ (speed = $\frac{\text{dist}}{\text{time}}$) $\quad \tau \downarrow \rightarrow \tau$ (Relaxation Time).

$I = n A e V_d$

CONDUCTIVITY :- distance $\rightarrow$ Insulators $>$ Semi-conductor $>$ Conductors
(h)
Conductivity $\rightarrow$ Conductors $>$ Semi-conductors $>$ Insulators
(σ)

Why Semi-Conductors ?

- $\rightarrow$ $\circ$ EASIER CONTROLLABILITY. (Readily Available).
- $\rightarrow$ $\circ$ LATTICE Structure.
- $\rightarrow$ $\circ$ Thermally generated current.
- $\rightarrow$ $\circ$ Cost Effective.
- $\rightarrow$ $\circ$ Abundance In Availability.

" $q = ne$ "

$i = \frac{dq}{dt} = \frac{d(ne)}{dt} = n \frac{dq}{dt} = \text{net}$

$n \uparrow \rightarrow i \uparrow$

TYPES OF SEMI-CONDUCTORS

INTRINSIC

→ pure form / Natural
 → These are NOT USED in VLSI Industry Mostly

Because they don't have good conductivity.

ELEMENTAL

Si, Ge

Group 14 elements

COMPOUND

GaAs, InP

EXTRINSIC

→ Intrinsic + DOPING !!

carriers / $e^- \uparrow \uparrow \uparrow$
 $I_d \uparrow \uparrow \uparrow$

N-TYPE

- Pentavalent impurities/dopants

- eg:- Sb, As, Bi, P
 Group 15 elements

P-TYPE

Trivalent impurities/dopants.

eg- Al, B.
 Group 13 elements

Need of doping ?? → To improve the conductivity of the device (MOSFET, BJT, JFET, pn junction)

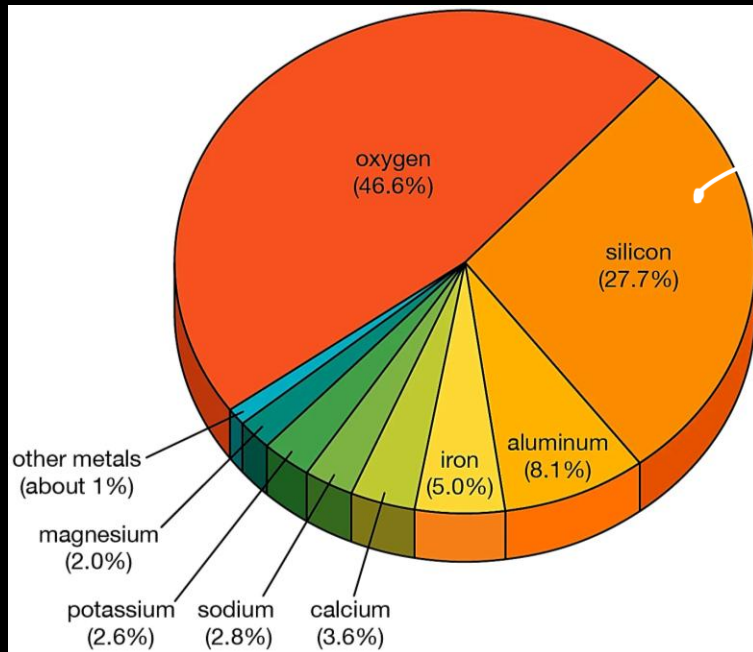
Physical Properties of Silicon

- 1 Its atomic number is 14 with a relative atomic mass of 28.085 u.
- 2 It has a metal-like appearance and is brittle.
- 3 Silicon's density is approximately 2.33 grams per cubic centimeter.
→ Thermally Stable ✓
- 4 Its melting point is 1410 degrees Celsius while its boiling point is 3265 degrees Celsius.
- 5 Silicon isotopes: -28, 29, and 30 are natural, but various artificial isotopes exist that are radioactive.
- 6 Its purest form is an intrinsic semiconductor.



Chemical Properties of Silicon

- 1 Its chemical behavior is purely electropositive. ✓
- 2 Its chemical behavior is very similar to metals.
- 3 It does not easily integrate with oxygen or other related elements because it is solid.
- 4 It is relatively inactive at room temperature.
- 5 At higher temperatures, silicon is highly reactive.
- 6 In a molten state, it forms alloys.



→ 2nd MOST ABUNDANT Mineral found in Earth's crust ...

CHIPS (I.C) → ~~Si~~ → defence → Glacier -4°C
-33°C
-20°C

Sahara 80-90°C, 62°C

MATERIALS BASED ON CONDUCTIVITY

